

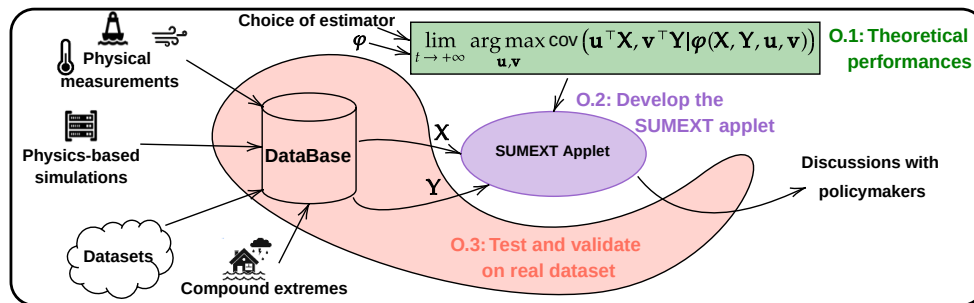
#@APP-FORM-HEMSCAPF@#

1 – Excellence #@REL-EVA-RE@#**1.1 – Quality and pertinence of the project’s research and innovation objectives** #@QUA-LIT-QL@#

Introduction. Extreme events can result in substantial damages and fatalities. During intense storms, strong winds can push water onshore, an extreme event denoted as storm surge. Due to sea-level rise, storm surges will certainly occur more often.¹ In association with heavy precipitation, they can trigger coastal floods, which directly threaten one third of the European population living within 50 km of the coast². Coastal floods are an instance of compound extremes (CExt), i.e., extreme events associated with at least two other extreme events. The compound structure of coastal floods can be studied by modelling jointly precipitation and storm surges^{3,4}. A recent work⁵ has showed an increase of the probability of flooding due to both precipitation and storm surges associated with climate change. They pointed out insufficient data to account for high dimensionality and insufficient flexibility in the modelled dependencies. CExts also occur in other areas, such as finance (e.g., the aftershock of the Lehman crisis) or wildfires (e.g., 4% of Europe’s total surface (almost 716kha⁶) has already burnt in 2022 due to combined fires). Facing this actual situation, it is needed to understand and prevent CExts. Those events cannot be reproduced at will, remains modelling. However: - storm surges are measured from buoys or simulated with physics-based model; - rainfall simulation models are not effective in predicting future periods. To circumvent this issue, we propose to model jointly storm surges and precipitation conditionally on meteorological (wind, pressure levels, temperature) and topographical (digital elevation model) variables. Such a bivariate regression model will allow to study the joint evolution of storm surges and precipitation through the evolution of meteorological variables which are better represented by climate models and are thus easily accessible.

The SUMEXT action proposes to 1) develop a new methodology able to model multiple extreme events based on multiple input variables even high dimensional; 2) implement those solutions in an open source applet; and 3) successfully apply it to dataset describing coastal flooding in Northeastern North America (NNA) and Europe.

The fellowship will permit such a project because of the mix of expertise of Prof. Julie Carreau, the outgoing supervisor (OS) at Polytechnique Montréal who has expertise in climatic and meteorologic multivariate modeling of extreme events; Stéphane Girard, the returning supervisor (RS) at the center Inria of the Grenoble Alpes University (UGA) who has expertise in extreme modeling and high dimensional context; and Hadrien Lorenzo, the postdoctoral fellow (PF) who has expertise in multiple response modeling and high dimensional context. Note that a two month academic secondment will be spent at the start of the project at Inria under the personal supervision of the RS. This will be an intensive course on advanced topics of extreme theory. This project outputs will feed into the EU Adaptation Strategy and to the European Green Deal offering, *in fine*, a tool to model the risk of CExts.



State-of-the-art. Supervised approaches of multiple-output description is a fairly old concept and the Ordinary Least Squares (OLS) solution has been the first proposition but it faced the curse of dimensionality as the empirical covariance matrix failed to be regular. H. Wold proposed the Partial Least Squares (PLS) method in the 60’s. PLS reduces the dimension of $\mathbf{X}$, a p -dimensional random vector of meteorological variables (such as wind, pressure and temperature), by looking for linear combinations aligned as much as possible with the answer $\mathbf{Y} = (Y_1, \dots, Y_q)^T$ (a q -dimensional random vector of potential extremes corresponding to the events that must be understood, such as the sea-level and the level of

¹Bouwer and Jonkman, “Global mortality from storm surges is decreasing”, 2018.

²From the *Adapting to rising coastal flood risk in the EU under climate change* by the European Commission Joint Research Centre.

³Zheng et al., “Modeling dependence between extreme rainfall and storm surge to estimate coastal flooding risk”, 2014.

⁴Zellou and Rahali, “Assessment of the joint impact of extreme rainfall and storm surge on the risk of flooding in a coastal area”, 2019.

⁵Bevacqua et al., “Higher probability of compound flooding from precipitation and storm surge in Europe under climate change”, 2019.

⁶From the European Forest Fire Information System (EFFIS <https://effis.jrc.ec.europa.eu>) a component of the Copernicus program.

precipitations) based on an algorithm designed in⁷. Note that vectors are in bold while scalars are not in the following and so $\mathbf{Y}$ can be denoted Y if $q = 1$. In order to deal with high dimensional cases ($\mathbf{X}$ and $\mathbf{Y}$ of large dimensions and strong correlation structure) the PLS approach has been widely adapted. Data-driven Sparse PLS (ddsPLS)⁸, developed by the PF, is mostly adapted to high dimensions. However this solution does not allow to predict extreme output. The PLS philosophy has been extended to the extreme context through the Extreme Partial-Least-Squares (EPLS)⁹, developed by the RS, solution in the case of 1-dimensional response Y . EPLS seeks linear combinations of $\mathbf{X}$ maximizing the covariance with Y given that Y exceeds a large value t which can be described by the maximization, over $\mathbf{v}$, of the following

$$\text{cov}(\mathbf{v}^\top \mathbf{X}, Y \mid Y > t), \quad (1)$$

where $\mathbf{v}$ is unitary. Also, $\text{cov}(\cdot)$, resp. $\|\cdot\|$, is the covariance function, resp. the Euclidean norm. EPLS does not support a multidimensional Y and so is not a correct answer to describe CExt events structure. The Principal Component Analysis for Multivariate Extremes (PCAME)¹⁰ assumes that some linear combinations of $\mathbf{Y}$ are more prone to extreme values than others. The authors study the maximization of the Euclidean norm of the multiple extreme $\mathbf{Y}$ normalized and projected onto a linear d -dimensional subspace spanned by the columns of a matrix $\mathbf{V}$ under condition that $\|\mathbf{Y}\| > t$. The authors study the asymptotic and non-asymptotic behaviors of their solution in order to provide error bounds in real cases where the data sample is of finite size. But this solution does not address the conditional aspect of the problem since the meteorological variable $\mathbf{X}$ is not taken into account. The OS and the RS have worked on a weighted likelihood solution in order to address jointly condition on $\mathbf{X}$ and low sample size¹¹, but this solution does not consider multiple extreme output. None of the previously discussed methodology takes into account the spatial structure of the data, meaning that values of meteorological variables geographically close to each other should also be close to each other. The approach SpatialXGumbel¹², developed by the OS, aims at estimating spatial extreme structure in a region taking into account the distance of the sampling sites (corresponding to different weather stations) from a reference. This solution needs just a few number of parameters to be tuned and is appealing for its interpretability. Another recent solution is evtGAN¹³ and relies on the convolutional neural network technology to take into account the spatial dependence of the data. This technology has great flexibility and performance, but suffers from poor interpretability and long learning times. While all the aforementioned approaches separately address some of the aspects of our present problem, none address all the constraints: ddsPLS does not consider multiple extremes, EPLS does not consider multidimensional $\mathbf{Y}$ and PCAME, SpatialXGumbel or evtGAN do not condition on the input covariate $\mathbf{X}$.

The ClimEx¹⁴ project led by the Ouranos¹⁵ consortium (localised in Montréal) and Environnement et Changement climatique Canada (ECCC), studies flood events that have occurred in Québec and Bavaria in order to propose solutions to decision makers to implement climate change adaptation strategies. To this end, a downscaling strategy is used where simulated data of resolution ≈ 310 km are enriched to ≈ 12 km resolution on the scale of the Europe and the NNA. Finally 50 replications (randomly initializing the model) of the previous downscaling step and covering the period of the 1950 to 2099, build the dataset. They project to use this first step work for studying the coastal floods in NNA and Bavaria. A slightly less precise ensemble of datasets, 25 to 50 km, and denoted CORDEX maps almost all the globe through fourteen regions. CORDEX has already been extensively studied, see¹⁶ or¹⁷ to analysis flooding events.

R&I objectives. To address the lacking aspects of the aforementioned approaches, this multiple-index inverse regression model is considered, with $\varepsilon \in \mathbb{R}^p$ independent from $\mathbf{Y}$, the link functions f_r are unknown:

$$\mathbf{X} = \sum_{r=1}^R f_r(\mathbf{Y}) \boldsymbol{\beta}_r + \varepsilon. \quad (2)$$

The $\boldsymbol{\beta}_r$ s are unitary, non-stochastic and not identifiable. Further, the subspace spanned by $\boldsymbol{\beta}_r$ is identifiable, see¹⁸ written by the PF and the RS. The PF will firstly study the parametric part of the previous model (the $\boldsymbol{\beta}_r$ s), assuming extreme

⁷Wold, “Estimation of principal components and related models by iterative least squares”, 1966.

⁸Lorenzo, Cloarec, et al., “Data-driven sparse partial least squares”, 2022.

⁹Bousebata, Enjolras, and Girard, “Extreme Partial Least-Squares”, 2021.

¹⁰Drees and Sabourin, “Principal component analysis for multivariate extremes”, 2021.

¹¹Carreau and Girard, “Spatial extreme quantile estimation using a weighted log-likelihood approach”, 2011.

¹²Carreau and Toulemonde, “Extra-parametrized extreme value copula: Extension to a spatial framework”, 2020.

¹³Boulaguiem et al., “Modeling and simulating spatial extremes by combining extreme value theory with generative adversarial networks”, 2022.

¹⁴Leduc et al., “The ClimEx Project: A 50-Member Ensemble of Climate Change Projections at 12-km Resolution over Europe and NNA”, 2019.

¹⁵See <https://www.ouranos.ca> for an overview of the consortium.

¹⁶McGinnis and Mearns, “Building a climate service for North America based on the NA-CORDEX data archive”, 2021.

¹⁷Di Sante, Coppola, and Giorgi, “Projections of river floods in Europe using EURO-CORDEX, CMIP5 and CMIP6 simulations”, 2021.

¹⁸Girard, Lorenzo, and Saracco, “Advanced topics in Sliced Inverse Regression”, 2022.

values of $\mathbf{Y}$, in objective 1 (O.1). The PF will address the non-parametric part in (O.3) in order to propose interpretable values through predictions of $\mathbf{Y}$. The PF identified two research objectives (O.1 and O.3) and one implementation objective (O.2). All will be detailed in Section 1.2 but are briefly described here.

Objective 1 (O.1): Derive theoretical performances. Define conditional multi-dimensional extremes and associated estimators, derive asymptotic and non-asymptotic performances and develop approximation algorithms for each estimator. The PF will consider *naive* estimators (WP2) and kernel-based estimators (WP3). Each solution will be challenged by in-depth simulations. This is feasible since it merges the EPLS and PCAME for which such results exist. Two research papers will measure the realization of this objective.

Objective 2 (O.2): Develop a user-friendly applet. Implement the estimation procedures in an applet. This applet must be optimised from an IT point of view. This is feasible since the PF has long expertise in R with 2 packages uploaded on the CRAN platform. R is a free software environment for statistical computing and data analysis oriented websites development. CRAN is a worldwide network that stores identical versions of code/documentation for R. A package updated on GitHub and on the CRAN will prove the measurability of this objective (WP4).

Objective 3 (O.3): Test and validate through real case datasets. Test and validate both the methodology and the implementation on the ClimEx and the CORDEX datasets associated with coastal flood risk analysis (WP6). The ClimEx dataset will be obtained in collaboration with Ouranos and ECCC (WP5). The CORDEX dataset will be obtained by downloading from the project's website. Finally the measurability of this objective relies on a publication.

1.2 – Soundness of the proposed methodology.

Overall methodology. Objective O.1 (WP2,WP3): The model (2) is most often studied index by index starting from the first one as we do here. Since $\mathbb{R}^q$ has no canonical ordered a real valued function φ is used to extend the notion of quantile to $\mathbb{R}^q$. The PF has decided to investigate the following solution behavior of

$$\arg\max_{\mathbf{v}, \mathbf{u}} \text{cov}(\mathbf{v}^\top \mathbf{Y}, \mathbf{u}^\top \mathbf{X} \mid \varphi(\mathbf{X}, \mathbf{Y}, \mathbf{u}, \mathbf{v}) > t) \quad \text{s.t. } \mathbf{v} \text{ and } \mathbf{u} \text{ are unitary,} \quad (3)$$

as $t \rightarrow \infty$, the extreme case. Four interesting functions φ are described in the table below with pros and cons

Form of φ	Explicit solution ?	Already studied ?	CExts ?	Extreme and high $ X_j $ s ?
$\sum_{j=1}^q Y_j^2 = \ \mathbf{Y}\ ^2$	Yes	Yes (PCAME)	No	No
$\sum_{j \neq k} Y_j Y_k $	Yes	No	Yes	No
$\sum_{(j,k)=(1,1)}^{(p,q)} X_j Y_k $	Yes	No	No	Yes
$ \mathbf{Y}^\top \mathbf{v} $	No	No	Yes	Yes

Only the fourth function takes both into account the compound structure (fourth column) and association with large values in $\mathbf{X}$ (fifth column) but this is also the single function which does not rely on an explicit solution. For any function φ , the solution must be associated with estimators in order to apply the proposed methodology to real datasets. Two types of estimators will be considered, firstly the *naive* estimators corresponding to empirical estimators of mean and variance operators (WP2). Secondly, a kernel-based estimator solution will be considered in close collaboration with the OS (WP3). This approach will allow to weight each observation according to a fixed kernel with regards to other observations. This kernel can be chosen in order to include the compound extremes structure (the $\mathbf{Y}_i$ s) aspect in the estimation of $(\mathbf{v}, \mathbf{u})$, or the covariate structure (the $\mathbf{X}_i$ s) for example. This approach introduces an extra-parameter which must be fixed. Cross-validation or bootstrap approaches will be considered by the PF, tools that the PF already used in the past. This step is computationally intensive, even after the coding has been optimised. Moreover, for each couple (estimator, φ), extensive simulations will be provided based on classical multivariate extreme distributions. The $\mathbf{u}_r$ s, for each function φ can be analysed within the spectrum of parameters β_r s in the inverse regression model (2). Indeed, space spanned $(\mathbf{u}_1, \dots, \mathbf{u}_R)$ is certainly (?) an estimator of the space spanned by $(\beta_1, \dots, \beta_r)$. The PF will derive theoretical results of asymptotic and non asymptotic behaviors of the naive estimators (i.e. asymptotic normality, error bound) in regards with the previous scientific question (WP2). For WP3, an interpretation of the kernel-based weighting will be sought with regards with the likelihood function deriving from Model (2) and the to be studied dataset. As the fourth proposed function does not have an explicit solution, there is a specific algorithmic challenge.

Objective O.2 (WP4): Based on the O.1 development code, the PF will develop an open-access and user-friendly web-interface: the SUMEXT applet. Since spatial structure data is considered in the application, the PF will adapt the solution developed by the OS (SpatialXGumbel) to SUMEXT. The R-package shiny will be used in the applet coding. Further the R-packages doParallel (allows parallel computations) and Rcpp (allows to embed fast codes in C++) will be extensively used to optimize the execution of the code. Note that those packages have already been widely used by the PF.

Objective O.3 (WP5,WP6): The PF will validate and demonstrate its approach on the ClimEx & CORDEX datasets (WP5). The first task of the PF will be to gather the different parts of the dataset. He will be helped by the OS who currently collaborates with the Ouranos consortium and ECCC. Data cleaning will draw on the PF's experience in dealing with missing data¹⁹ and outlier detection²⁰. The resulting database (DB) will then be analysed according to the previous methodology (WP6). In order to assess for sensibility in the finite size sample context, extensive re-sampling methods (such as cross-validation or bootstrap) will be considered. Thus, the stability of the subspace spanned by $(\mathbf{v}, \mathbf{u})$ will be monitored. Under model (2), kernel-smooth estimator will allow non-parametric estimation of $\mathbf{Y}$, noted $\hat{\mathbf{Y}}$, based on $\mathbf{u}^t \mathbf{X}$. The precision of the developed model can therefore be controlled through estimation of $|Y_k - \hat{Y}_k|$ and allows to validate the approach on ClimEx and CORDEX datasets. Also, discussions with climate scientists and meteorologists who collaborate with the OS will allow to validate the results. The workplan is described in detail in 3.1.

Integration of methods and disciplines. This action depends on the development of a statistical framework and associated estimators in order to be implemented, it mixes in those parts statistics, probabilities, algorithmic, climate science, software and web development, data shaping and management and data analysis skills. In total, six disciplines are at the core of this action, the implication of each work package in those disciplines is described in this table:

	Probability & Statistics	Algorithmic	Climate Science	Software & Web Dev.	Data shaping	Data analysis
WP2: <i>Naive estimators</i>	• • •	• •				
WP3: <i>Kernel estimators</i>	• •	• • •				
WP4: <i>SUMEXT applet</i>		•	• •	• •		
WP5: <i>Build DB</i>			•		• • • •	
WP6: <i>Analyse DB</i>			• •			• • •

The number of • corresponds to the implication of each discipline in each WP from no implication ($0 \times \bullet$) to maximal implication ($5 \times \bullet$).

Gender dimension. No gender dimension is foreseen since the project deals with mathematical and climatic concepts.

Open science practices and data management. Open sharing of pre-prints, articles, data sets and software will be performed through publishing on open-access repositories such as *hal.inria.fr*, *arxiv.com*, *figshare.com* and *github.com* respectively. In order to detail the functionalities of the developed software, step-by-step data manipulations and output interpretations scripts denoted as vignettes will be written and posted on the CRAN page of the package. A specific vignette will be developed for the case of environmental data. The NetCDF (NETwork Common Data Form) format is a binary format suitable for storing multidimensional meteorological variables. Thus, the previous vignette must explain how to deal with this format. Further, SUMEXT must be applicable to other fields than environment, such as finance and so a vignette must be general enough to match with all communities. Finally, the PF will participate in a workshop/hackaton (such as <https://launchlab.ai/fr/climate/>) to include third party actors in the development of the project. In addition, once the applet is completed, an open access publication will be produced to explain its overall functionality. All these aspects ensure open science.

The DB will be stored on *figshare.com*, a solution which maintains FAIR data principles and allows storing NetCDF-files. In particular, all data are assigned to an unique Datacite DOI and rich metadata and are indexed in Google Dataset Search which proves the *Findable* aspect of this solution. No account is mandatory to access the publicly published data. Further, data will never be hidden by a paywall. Also, in the case of sensible data, that cannot be publicly released, metadata is still visible. Finally publisher portals can access the data via <https> protocol. All these aspects make this solution *Accessible*. Metadata can be written in a large variety of languages such as [html](https) which will be favored by the PF. Also, this solution can be linked to peer reviewed publication and cite any other dataset which insures the *Interoperable* aspect. Each repository can be associated to a *github.com* project which will simplify the writing of the metadata which will allow the future user to precisely understand which operations have been performed to generate datasets. All members of the climate and extreme communities must be able to find their standards in this repository. The dataset will be released under CC0 licence. For these reasons, *Reusable* aspect is met. Note that the PF already dealt with a FAIR dataset uploading in

¹⁹Cloarec, Lorenzo, and Saracco, "Koh-Lanta, missing data imputation in supervised context", 2022.

²⁰Lorenzo and Saracco, "Computational Outlier Detection Methods in Sliced Inverse Regression", 2021.

the case of a sensible dataset for Ebola vaccine trial²¹ for which an anonymisation solution had to be put in place. The data management plan (DMP) will be delivered at month 6 and will match the dissemination plan detailed in Section 2.2.

1.3 – *Quality of the supervision, training and of the two-way transfer of knowledge between the PF and hosts.*

Supervision. Julie Carreau (Polytechnique Montréal, OS) is Assistant Professor in the Department of Mathematical and Industrial Engineering (2021-) and is also affiliated with IVADO, the Institute for Data VALORIZATION. She has a mixed background in mathematics and machine learning. She has spent 14 years in France working as a research scientist, initially as a postdoctoral fellow and then with a tenure position at *Institut de Recherche pour le Développement* (IRD). As a postdoctoral fellow, she stayed at Inria GRA for about six months working with the RS on a topic linked to the SUMEXT proposal, see reference¹¹ page 2. During her 11 years at IRD, the OS worked mostly on the modeling of extreme events in the context of climate and hydrological sciences and more particularly to study heavy rains and floods. She spent two years (from 2019 to 2021) in Tunis (Tunisia) in the framework of the project Amande (Hubert-Curien UTIQUE Programme), of which she was a coordinator. She also participated in many French national projects such as Fraise or the ANR project StaRMIP 2013-2017. Since back in Montreal, the OS obtained a joint starting grant from Polytechnique Montreal and IVADO and the Discovery Grant from NSERC-CRSNG to fund her research program on the stochastic simulation of high resolution data for flood risk assessment. She is serving as associate editor for the Journal "Stochastic Environmental Research and Risk Assessment" (2022-2025). She has published about 40 articles in statistical journals such as *Spatial Statistics* and *Extremes* and applied journals such as *Water Resources Research* and *Advances in Water Resources*. She has been involved in the co-supervision of four Ph.D. students (one of which is currently a postdoc at IRD) and two postdocs. Stéphane Girard (Inria GRA, RS) is Research Director (1st grade, DR1) in the STATIFY team and, from 2018, part-time associate researcher at Ecole Polytechnique. He is/was associate editor of *Statistics and Computing* (2012–2019), *Revstat-Statistical Journal* (2019–) and *Journal of Multivariate Analysis* (2016–). He is also member of the advisory board of *Dependence Modeling* (2015–). He was head of the statistical department (≈ 35 permanent members) of the Jean Kuntzmann Laboratory - LJK (2012–2016) at the Université Grenoble-Alpes (UGA) and head of the associate team Simerge joint between Inria Grenoble Rhône-Alpes and Université Gaston Berger, Saint-Louis, Senegal (2015–2021). He is a specialist of extreme-value theory and high-dimensional statistics. Stéphane Girard published more than 120 research papers in top-ranked journals including *Annals of Statistics*, *Journal of Machine Learning Research*, *Bernoulli*, *JRSS-B*, with 5000+ citations and h-index 40 from google scholar. He has been co-supervising 20 PhD students (3 of them are now full professors) and 10 postdocs thanks to several M€ fundings obtained from Inria, UGA, French Agency of Research (ANR), among others. He was co-organiser of a series of four astrostatistics schools (2013–2019) bringing together astrophysics students and statistic teachers, the courses being published in textbooks by EDP Sciences. He also co-organised three editions of the annual workshop Statlearn (12 invited international speakers).

Planned training activities for the researcher. The secondment will take place at the start of the action and will last two months. It will treat of advanced topics of extreme value theory and will be provided by the RS and by training-through-research. During the outgoing phase, training-through-research will be considered to transfer skills and knowledge about two different points. First, the OS and the PF will directly interact to dimension interesting estimators and to adapt the SUMEXT approach to spatial data context. Second, the PF will learn on the NetCDF format on the basis of references given by the OS. During Autumn 2023, the PF will attend an advanced course on extreme value analysis given by Debbie Dupuis, a world-renowned researcher on extreme values as applied to climate change. This course is given at HEC Montreal and is accompanied by financial use cases, thus opening new fields of application. The PF will also participate to the course Leading a research project proposed at the outgoing host institution to PhD students. He will also attend to the DB oriented course *Fichiers et bases de données* ($\approx$ Files and databases) from Polytechnique Montréal. In order to improve his skills in student supervision, the PF will co-supervise a student (whether PhD or BSc) with the OS and the RS. The choice of transversal formations chosen by the PF matches with his personal ambitions since he indeed wants to improve his skill communicating in English both oral and written. This aspect will be completed by a transversal formation on open access and open data. Also, he will work on professional development in order to maximise its chances of obtaining a permanent research position.

Training activity	Training level	Scientific (S) or transversal (T) skill	Time
Extreme theory	RS	S: Training-through-research	2 months: start of WP2

²¹Rechtien et al., "Systems Vaccinology Identifies an Early Innate Immune Signature to the Ebola Vaccine rVSV-ZEBOV", 2017.

Estimator propositions & spatial adaptation	OS	S: Training-through-research	10 days: WP2, start of outgoing phase
NetCDF format & large dataset	OS	S: Training-through-research	5 days: start of WP3
Files and databases	OHI	S: Course	5 days as available
Supervise undergraduate or PhD students	OHI/RHI	S: Training-through-research	2 days as available
Analysis of extreme values with application in financial engineering	OHI (HEC Montréal)	S: Course	2 days as available
Leading a research project	OHI	T: Course	2 days as available
Open data & open access	RHI	T: Course	1 day as available
Scientific communication & writing	RHI	T: Course	2 days as available
Obtain a permanent research position	RHI	T: Course	2 days as available in returning phase
Project, grant management and applications	OS & RS	T: Project management and grant writing, objective ERC	1hr/month, throughout all project (WP1)

Acronyms: OHI for outgoing host institution; RHI for returning host institution.

Three-way transfer of knowledge between the researcher, host organisation, and associated partner for outgoing phase. *From supervisors and host institutions to the PF.* As detailed in the previous table and the associated paragraph, the supervisors and associated institutions will support PF activities in training scientific and transversal skills, via regular meetings and specific courses, and also contribute to extend his network. Also, hosting arrangements detailed in 3.1 demonstrate this aspect.

From the PF to host and between institutions. The PF has expertise on many scientific subjects such as high dimensional analysis, variable selection, missing value processing, dimension reduction, R programming. He will share his knowledge through seminars or short courses at IVADO or related research labs thanks to the OS, member of this organization. The PF will participate to the supervision of a undergraduate or PhD student, shared also with the RS and the OS. Thus, the PF will directly contribute to the research of the OS and the RS. The PF and the pair RS-OS will organize a session at CMStatistics on the double topic of extremes and environment. During the outgoing phase, he will participate to European conferences and will use this opportunity to spend one week at Inria and work with the RS. Also, the RS will spend a week in the OHI during the outgoing phase and the OS will spend a week in the RHI during the returning phase. These stays will enable the PF, OS and RS to work more effectively and thus to increase the chance to address all the objectives of the SUMEXT project.

1.4 – Quality and appropriateness of the researcher’s professional experience, competences and skills

The researcher has extensively worked on the PLS model during his PhD and his postdoctoral position (for the past two years). This approach is at the core of the EPLS solution and the present SUMEXT project. Further, the PF has already published a paper, with the RS, in the *Journal of MultiVariate Analysis (JMVA)* one of the top journals in statistics and he is currently working on another with the RS on the topics of variable selection, dimension reduction and non linear supervised approach. The PF already interacted with members of the team STATIFY (in which the RS is a member) especially Julyan Arbela young researcher and discussed about mix of extreme theory and Bayesian approaches. This proves the capacity of the PF and the RS to work together and to be well-integrated in the Inria team STATIFY during the returning phase. The SUMEXT project needs flexibility and adaptation because it merges many different aspects of statistics, coding and climatic modelling. The PF has also developed many open access software (two of which have been put on the CRAN) and he had to deal with dataset access under the FAIR concept in the case of sensible data (Ebola vaccine data). These aspects cover more than two-third of the overall project time. The PF is used to learn new methods and tools (as he did for application projects to vaccine research or thromboembolic research) which will be needed for this project. For this he will both train-through-research and participate to courses in the host institutions on both of extreme value theory and climatic modelling. Transferable skills of the PF are numerous and he developed national and international collaborations (twelve co-authors, ten national and two international). He also co-supervised four internships and participated to the steering committee of a thesis. He is currently working with a former intern now at UCLA on a research paper on variable selection and dimension reduction, project that has already been implemented in a R package and released on the CRAN. He was also twice member of the Jury of an university degree entitled *Big Data and Statistics for Engineers*. He contributed to different dissemination events such as *La fête de la Science* ($\approx$ The Science Festival) or Dataquittaine..

2 – Impact #@IMP-ACT-IA@#

2.1 – Credibility of the measures to enhance the career perspectives of the researcher and contribution to his skills development

After this fellowship, the PF will apply for permanent positions in French academic institutions such as Inria and CNRS. The PF wants to work on dimension reduction associated question, both through theoretical and practical ways. The PF developed an overview of problems associated with high dimensional data and the SUMEXT project will take this profile even further through three aspects. Firstly this action will consolidate PF research existing skills (see 1.4) reinforcing them thanks to variate interdisciplinary activities: statistical and extreme theory; data structures, formats and diffusion (see 1.3). This will be performed thanks to extensive use of training-through-research in contacts with top researchers: the OS, the RS and its contacts in the Ouranos consortium, the ClimEx project and the IVADO institute but also in contact with promising students. Secondly, the PF will acquire and develop soft skills related to the transversal topics of: supervision experience, communication and writing for research and grants but also data policies trough open-access and open-data. The objective of the PF is to write an ERC Starting Grant by the end of the project, making use of the RHI internal resources, as permitted by the fellowship. Thirdly, PF's global visibility will benefit from the fellowship working with renewed researchers and scientists. Collaborating with Ouranos consortium and IVADO institute will certainly bring other collaboration opportunities and grow the PF's professional network. For example, IVADO is *An Artificial Intelligence Research and Transfer Institute* and gathers Montréal top actors in this subject, this is an ideal location to discuss the proposed solution (see Equation (3) in 1.2) and imagine new collaborations through different approaches. Obtaining renowned funding such as MSCA PF will boost the PF profile faster than any other type of temporary position.

2.2 – Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities #@COM-DIS-VIS-CDV@#

Plan for the dissemination, including communication activities. The dissemination strategy was adopted in order to maximise the project impact towards well-defined targets and will be further detailed in our Dissemination and Exploitation plan (D7.1). In terms of scientific productions, four research papers will be produced in journals and five conferences will be attended with associated proceedings. The participation to a workshop/hackaton will help showcase important concepts and results from the project to third parties, possibly including other domains of interest (other areas of environmental sciences, epidemiology, finance)

Plan for the exploitation activities. It is not expected that any commercially exploitable intellectual property will be produced during this fellowship. However, exploratory measures will be taken to look for longer term transfer possibilities (companies, public bodies and meteorological agencies) via the RHI's technology transfer office. According to 1.2, data and codes will be made public under the CC0 license in a FAIR framework improving peer-review, potential public acknowledgement and references by the scientific communities. The PF will approach the *Observatoire National des Risques Naturels* (ONRN, $\approx$ National Natural Hazards Observatory) to collaborate on the understanding of CExtS and the creation of a risk index associated with the occurrence of CExt events. This will directly exploit the results of the SUMEXT project. Dissemination, exploitation and communication activities are detailed below

Target	Message	Tool	Channel
Communication			
General public	Inform about maths & statistics research, extreme events, climatic modelling	Events	Pint of Science (RHI, 1/year in April, 30-50👤), European Researchers' Night (France, 1/year in September, 1000🎓), <i>La journée des Lycéens</i> (UGA, 1/year in January, > 7000👤), <i>La fête de la Science</i> (France, 1/year in October, > 1million👤)
		Online	Personal page, ACFAS, QuébecScience ([3, 5]10 ³ characters)
		Call for proposals	<i>Tous résilients face aux risques</i> : A call by the French government every year for mid-October: events to prepare for climatic disasters.
Dissemination			
Statisticians	Description of the statistical approach	Journal	JMVA _{1.471} ; Extremes _{1.255} , Annals of Applied Statistics _{2.083} , Statistics and Computing _{2.559}
		Conference	EVA ₃₀₀ , CMStatistics ₁₉₀₀ , JDS

Climatologists	Application to DB	Seminar	IVADO: on-demand seminars thanks to the OS
		Journals	Journal of Hydrology ^{6,708} , Water Resources Research ^{5,711}
		Conference	EVAN, IMSC
R users	Overview of the applet	Seminar	Ouranos consortium: on-demand seminars thanks to the OS
		Journal	JSS ^{6,274} , Environment Modelling & Software ^{5,288}
Academic	Teaching the approach to students	Course materials	ED MSTII: UGA doctoral school of Mathematics, Information Science and Technology, Computer Science. Already experienced by the RS for a course on extreme value theory.
	Co-development	hackaton workshop	Organised by piggybacking on attended conferences and seminars
Exploitation			
Policymaker	Propose SUMEXT approach	Seminar	ONRN: an institution associated with the French government. Approach to creating a risk indicator of CExt events. Alternatively the Canadian Climate Services Centre (CCSC).

Journals subscripts give the impact factor and conference subscripts give the audience. **Symbols & acronyms:** : public attendees; : researchers; UGA: *Université Grenoble Alpes*; JMVA: Journal of MultiVariate Analysis; EVA: Extreme Value Analysis; CMStatistics: Computational and Methodological Statistics; JDS: *Statistics Days*; Days: Statistics; EVAN: Extreme Value Analysis and Application to Natural Hazards; IMSC: International Meeting on Statistical Climatology; JSS: Journal of Statistical Software.

2.3 – The magnitude and importance of the project’s contribution to the expected scientific, societal and economic impacts

Expected scientific impacts. The SUMEXT project will impact the way scientists do CExt analyses by providing a tool that did not exist: perform predictions in a general, supervised, high dimensional framework. The first community impacted will be the one studying studying coastal flood, $\simeq 50$ teams worldwide. However since other environmental or financial domains can make use of the approach the PF will develop, and we have measures to foster usage in other domains, larger communities will be impacted. In particular, scientists working on other kinds of environmental CExt such as (wildfires+rainfall)²², (tropical cyclone+heat wave)²³ and financial risk analysis will be impacted. Similarly, other data structures exist and adaptation to the longitudinal context can be imagined taking into account climate change in coastal flood risk²⁴. Furthermore, new collaborations might emerge adapting the spatial dependence structure through convolutional neural networks such as done in evtGAN (see 1.1). Finally, the ONRN institution gathers database detailing other extreme events such as avalanches or wildfires. Thus, new scientific common interests can appear in parallel to the exploitation by ONRN of the SUMEXT project.

Expected societal impacts. The SUMEXT project will allow better prediction of catastrophic events such as coastal floods, reducing the tremendous damage they infer on urbanised coastal areas. Through awareness rising to both public bodies and general public (see 2.2), the project will also foster faster public and individual measures for risk prevention (see exploitation measures with ONRN or CCSC in 2.2). Today, half of the world’s population lives within 150 km of the coastline. By 2035, 75% of the world’s population will live in this area. This entire population could benefit from the results of the SUMEXT project. CExt negative effects meet with UN SDGs such as: endangering crops and livestock (Goal 2); deterioration of sanitary conditions facilitating the emergence of epidemics (Goal 3); destruction of school buildings (Goal 4); contamination of water sources with pathogens or agricultural chemicals (Goal 6).

Expected economic impacts. According to²⁵, 20% of world’s infrastructures will be endangered by coastal flooding by 2100, which represents \$14.2 trillion worldwide. To take a particular example, the flood of November 1994 in Nice (France) has lasted only 72 hours. Due to the high slopes behind the city and buildings constructed on flood-prone land, the Var river overflowed its banks and flooded an active part of the city (airport, shopping centre). The loss is now estimated at €20 million. Today, the *Observatoire Régional des Risques Majeurs* (ORRM $\approx$ Regional Observatory of Major Risks) has fixed three objectives and the main one is to better forecasting of rapid flooding of the river and valleys. In magnitude, our solution can be applied after consolidation by weather agencies in Europe and in Canada.

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²²Touma et al., “Climate change increases risk of extreme rainfall following wildfire in the western United States”, 2022.

²³Matthews, Wilby, and Murphy, “An emerging tropical cyclone–deadly heat compound hazard”, 2019.

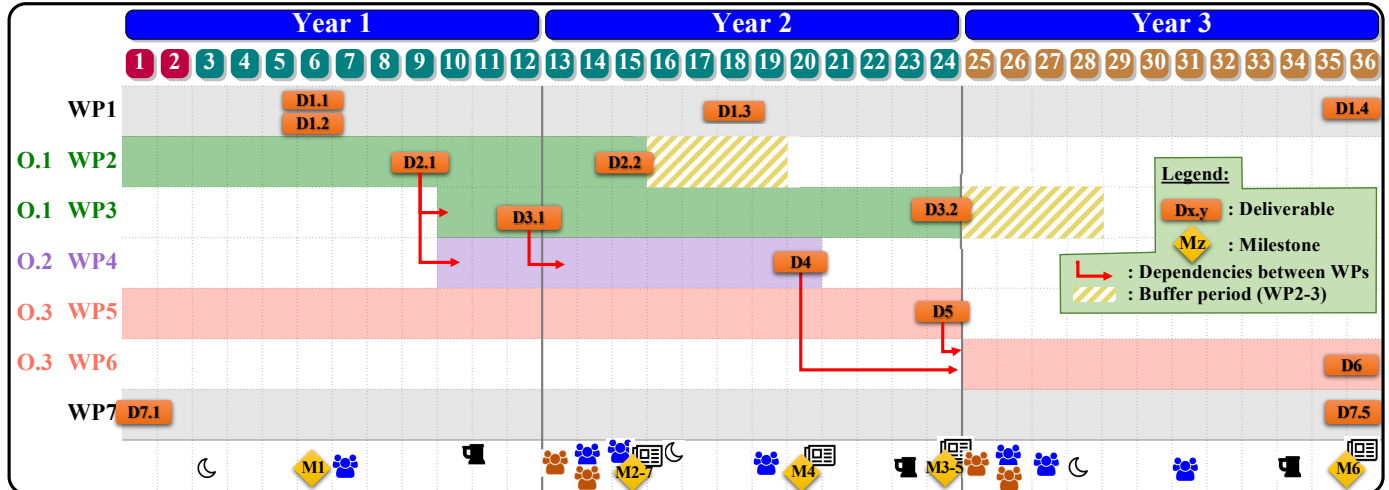
²⁴Bevacqua et al., “Higher probability of compound flooding from precipitation and storm surge in Europe under climate change”, 2019.

²⁵Kirezci et al., “Projections of global-scale extreme sea levels and resulting episodic coastal flooding over the 21st Century”, 2020.

3 – Implementation #@WRK-PLA-WP@##@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.1 – Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

Structure of the workplan. The following Gantt chart and the table detail and organise the WPs, tasks, deliverables and milestones. WP2 and WP3 relate to the development of the approach and its statistical properties for which buffer periods are proposed in order to mitigate potential difficulties. WP4 relates to the development of the applet and only needs partial realizations of WP2 and WP3. WP5's duration is fixed to two years, long enough to mitigate any trouble getting the data. Finally WP6 needs both WP4 and WP5 to be realized. The expected start of the activities is June 2023. Note a dependency of WP3 from WP2 which corresponds to the approximation algorithm chosen for complex functions φ such as $|\mathbf{v}^\top \mathbf{Y}|$. Also, WP4 depends on D2.1 and D3.1 which are the implementations of the chosen approximation algorithms for naive estimators and kernel estimators.



Second row corresponds to months (secondment, outgoing phase, returning phase); Dissemination: 📄: publication; 🧑: statistical conference; 🌿: environmental conference. Communication: 📢: Pint of Science; 🌙: European Researchers' Night.

Work package	Aims, Tasks (T), Deliverables (D) and Milestones (M)
WP1 Project management (PF: 3pm, OS+RS: 0.25pm)	Aims Planning, monitoring and evaluation of the project. T1.1 2/month progress checks with supervisors; T1.2 4/year meetings with Ouranos; T1.3 2/year CDP meeting; D1.1 CDP; D1.2 DMP; D1.3/1.4 mid-term and final project reports to EU. M1 CDP & DMP completed.
WP2 Naive estimators (PF: 7pm, OS+RS: 0.75pm)	Aims Derive estimators properties + code + simulations. T2.1 Attend secondment; T2.2 design estimators + approximation algorithms; T2.3 implement the approximation algorithms; T2.4 perform simulations; T2.5 derive theoretical performances. D2.1 Codes for estimations; D2.2 Theoretical property proofs & simulations. M2 Estimators understood + OA dev. code completed.
WP3 Kernel estimators (PF: 7pm, OS+RS: 0.5pm)	Aims Derive approximation algorithm properties + codes + simulations. T3.1 design estimators & approximation algorithms; T3.2 implement the approximation algorithms; T3.3 perform simulations; T3.4 design procedure to fix kernel-parameters. D3.1 Codes for approximation algorithms; D3.2 Theoretical property proofs & simulations. M3 Estimators understood + OA dev. code completed.
WP4 SUMEXT applet (PF: 7pm, OS+RS: 0.5pm)	Aims Fully embed the development code of SUMEXT in a Shiny applet, optimizations through low-level code inclusion and parallelization. T4.1 adaptation to spatial structure; T4.2 embed the development code of SUMEXT in an applet. D4 SUMEXT applet + OA. M4 SUMEXT applet available.
WP5 Build DB (PF: 3pm, OS+RS: 0.25pm)	Aims Build the database. T5.1 Gather data from the different sources; T5.2 handle spatial structures of the data (NetCDF) and dimension with regards to size of data (large datasets); T5.3 clean data; T5.4 merge data. D5 Open-access database on <i>figshare.com</i> . M5 DB built.
WP6 Analyse DB (PF: 4pm, OS+RS: 0.25pm)	Aims Test and validate the approach on coastal flood risk through DB. T6 Test & validate the SUMEXT models and the SUMEXT applet on DB. D6 OA analysis + validation results. M6 Approach tested and validated on DB.
WP7 Training, communication, dissemination, exploitation (PF: 5pm, OS+RS: 0.5pm)	Aims Training and impact-generating activities. T7.1 Prepare communication materials; T7.2 attend conferences; T7.3 prepare dissemination seminar (Ouranos); T7.4 organise exploitation activities targeting academics, policymakers and general public, see 2.2; T7.5 organize a CMStatistics session. D7.1 Dissemination and Exploitation plan; D7.2-4 papers; D7.5 exploitation activities (report on CMStatistics session, on meeting with ONRN). M7 Dissemination and Exploitation plans completed.

Acronyms : pm for person-months; OA for Open Access; CDP for Career development plan, DMP for Data management plan

Risk assessment. The following table describes the risk associated with each WP, their probability of occurrence (**P**), the impact levels (**I**) and mitigation measures.

Risk description, probability (P), impact (I)	Risk mitigation measures
No approximation algorithms found associated with the fourth function $\varphi : \mathbf{u}^\top \mathbf{Y} $ (WP2 & WP3), P=low, I=high	Simplify the chosen estimator in order to define an algorithm based on a linear optimization problem. Due to the form of the solution found in 1.2, the naive estimator corresponds to an eigenvalue decomposition and so can be chosen. 3 months can be chosen as a dumper to resolve this issue.
Difficulties in deriving theoretical properties in the non-asymptotic case mainly: the error bound (WP2), P=medium, I=low	Theoretical aspects are already treated for EPLS and PCAME for particular cases and naive estimator. SUMEXT is a generalization and proofs will adapt to functions 1 to 4. A problem can appear for the fourth function (no explicit form) and the 3-months buffer period can be used to solve this.
Lack of computational resources to fix the hyper-parameters of the kernel (mainly) in simulations (WP3), P=low, I=high	Hosts provide access and formation for means of calculation: Inria (e.g. Plafrim, JeanZay), Polytechnique Montréal (e.g. CalculQuébec). Here, the buffer period can be used to obtain a formation to handle means of calculation.
Data not shared (WP5), P=low, I=medium Data cannot be made freely available due to sensible data (WP5), P=low, I=medium	If ClimEx is not shared from Ouranos. The PF will use CORDEX solely. <i>figshare.com</i> offers solutions to host sensible data under FAIR approach. Showing only the metadata is part of their solution.
Exploitation avenue with ONRN not fulfilled (WP7), P=medium, I=medium	Approach weather agency such as Météo-France or the Canadian Climate Services Centre (CCSC), created to help Canadians plan for the future in light of climate change impacts.

3.2 – *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

Polytechnique Montréal and Julie Carreau’s lab (outgoing host): Founded in 1873, Polytechnique Montréal has educated over 52500 engineers. It ranks first in Québec in terms of the scope of its research activities. Its annual research budget stands at \$110 million and its 60 research units allow its teams of professors. Polytechnique accounts for the production of nearly one-quarter of university engineering research in Québec.

PM will give the PF help in installation, finding accommodation and integration of the family and spouse. PM will provide a desk (if available) and free access to computer resources at PM for the duration of his stay. Also, PM will provide PF means of calculation and courses to handle those means (due to particular languages such as MPI, OpenMP, CUDA, OpenACC).

The lab of Julie Carreau (OS) is quite recent since she took her function last year. She is an experienced researcher, see 1.3, benefiting from the dynamics of important education institutions in Montréal: Polytechnique Montréal, McGill, HEC Montréal. The PF will train on different subjects thanks to these institutions. The OS is a member of the IVADO institution, especially dedicated to knowledge transfer in artificial intelligence in Montréal. This proximity will allow the PF to give seminars on its research subjects to artificial intelligence actors of Montréal. The OS also collaborates with Prof. Jonathan Jalbert, expert in statistical climatology and long time collaborator of the Ouranos consortium. As a result, the PF will benefit from the views of climate experts.

The PF will participate to the supervision of master students shared with the RS during this period.

Inria and the STATIFY team (beneficiary & returning host): Inria was awarded the European Union’s HRS4R label in 2019, recognising its commitment to provide the best environment for its researchers. Inria will provide the PF an office and free access to computer resources at Inria for the duration of his stay. Also, the PF will have access to courses and training in Inria and more generally in UGA. In order to handle the different means of calculation available at Inria (such as JeanZay or Plafrim), the PF will have access to courses and can ask for research engineers to aid in the development of numerical tools. More particularly, the GRICAD service of UGA will provide technical support.

The PF will be integrated in the STATIFY team, headed by Florence Forbes and built on five permanent researchers, three research directors and two research scientists. This integration will occur through an introduction seminar and by attending to the lab meetings. Moreover, the PF will have the opportunity to participate to the life of the team through supervision of master students. The PF will have the opportunity to spend a few days there, before the beginning of the SUMEXT project, in order to finish an ongoing work with the RS and to meet the members of the team.

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